



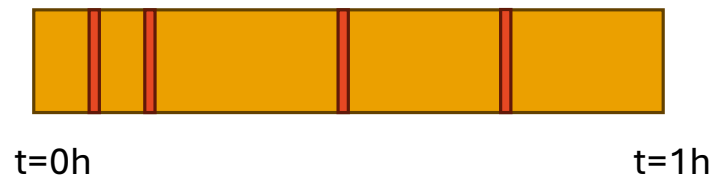
4 – Cont. Random Variables

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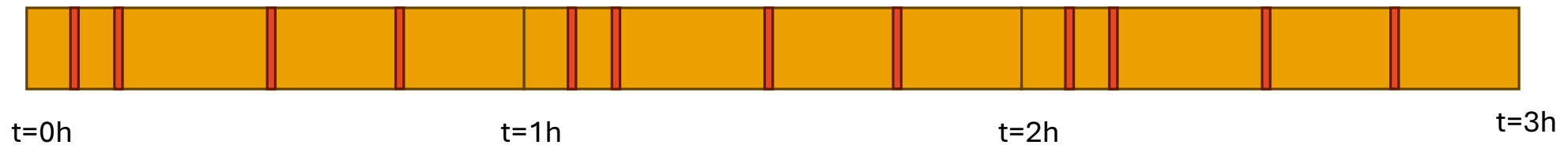
What is Poisson?

$$X \sim \text{Poisson}(\lambda)$$

↑
of occurrences per **unit time/distance**



$X \sim \text{Poisson}(4 \text{ per hour})$



$X \sim \text{Poisson}(12 \text{ per } 3 \text{ hours})$

$X \sim \text{Poisson}(4t \text{ per } t \text{ hours})$

A discrete random variable X has a **Poisson distribution**, denoted by $X \sim \text{Poisson}(\lambda)$, if its PMF is of the form

$$p(x) = e^{-\lambda} \frac{\lambda^x}{x!}, \quad x = 0, 1, 2, \dots,$$

where $\lambda > 0$.

Theorem 1 (Poisson)

If $X \sim \text{Poisson}(\lambda)$, then

$$\mathbb{E}[X] = \lambda, \quad \text{Var}[X] = \lambda.$$

Problem 1 (Poisson distribution)

- (a) In the past, computer tape was used to store files, and flaws occurred on these tapes. Suppose that flaws occur on the average of one flaw per 1200 feet. Let X be the number of flaws in a 4800-foot tape.
- (i) What is the PMF of X ?
 - (ii) What is the probability that $2 \leq X \leq 4$?
- (b) Let X have a Poisson distribution such that $3\mathbb{P}(X = 1) = \mathbb{P}(X = 2)$. Find $\mathbb{P}(X = 4)$.

Recap: PMF and CDF

The **probability density function (pdf)**, denoted f , of a continuous random variable X satisfies the following:

1. $f(x) \geq 0$, for all $x \in \mathbb{R}$

2. f is piecewise continuous

3.
$$\int_{-\infty}^{\infty} f(x) dx = 1$$

4.
$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Relationship between PDF and CDF for a Continuous Random Variable

Let X be a continuous random variable with pdf f and cdf F .

- By definition, the cdf is found by *integrating* the pdf:

$$F(x) = \int_{-\infty}^x f(t) dt$$

- By the Fundamental Theorem of Calculus, the pdf can be found by *differentiating* the cdf:

$$f(x) = \frac{d}{dx} [F(x)]$$

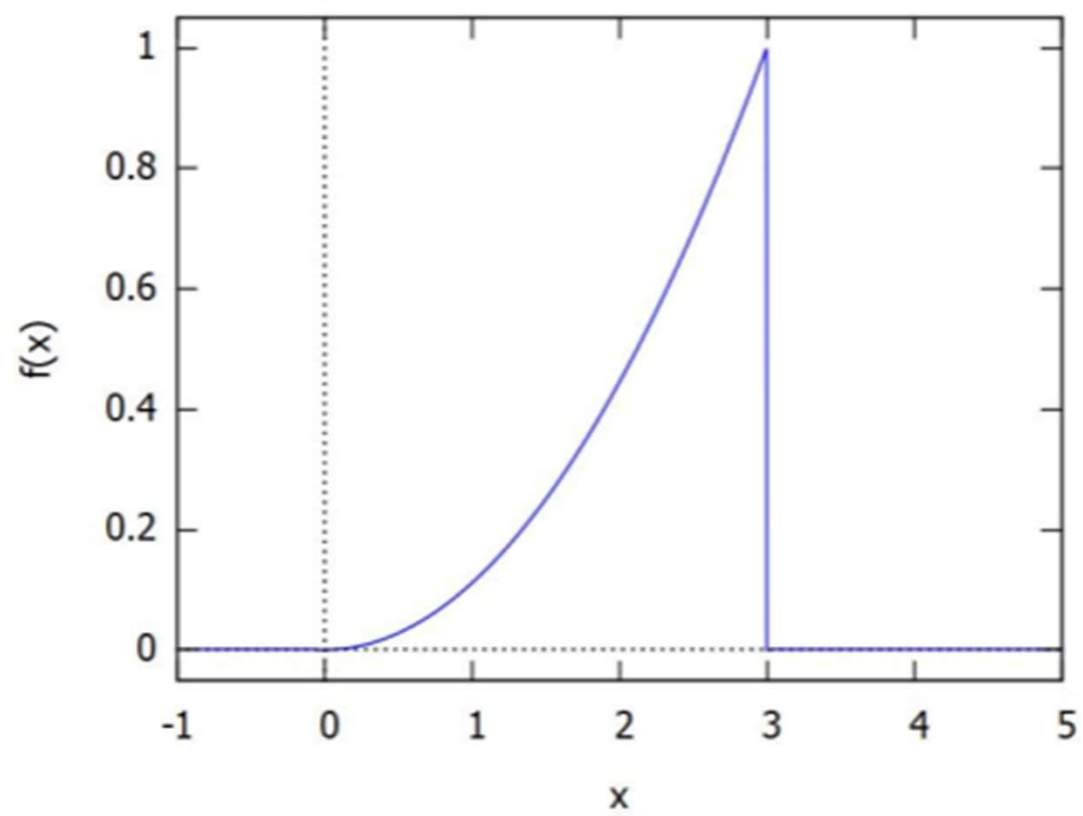
Problem 2 (Continuous random variables, PDF and CDF)

Let X be a random variable with PDF given by

$$f(x) = \frac{x^2}{9} \text{ for } 0 \leq x \leq 3 \text{ and } f(x) = 0 \text{ otherwise.}$$

- (a) Draw a graph of f .
- (b) Compute the CDF F of X .
- (c) Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\text{Var}[X]$.
- (d) Find $\mathbb{P}(X \geq 1)$, $\mathbb{P}(2 \leq X \leq 3)$, and $\mathbb{P}(X^2 \leq 2)$.

(a) Plot of the PDF $f(x)$:



Problem 3 (Continuous random variables, PDF and CDF)

Suppose X is a random variable whose PDF is defined by

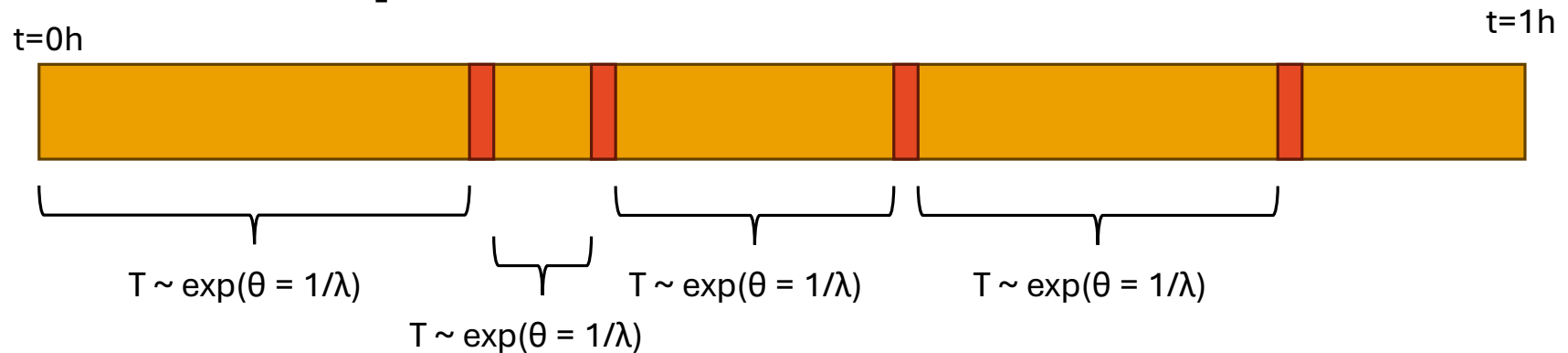
$$f(x) = \begin{cases} x, & 0 \leq x < 1, \\ \frac{c}{x^3}, & 1 \leq x < \infty, \\ 0, & \text{elsewhere.} \end{cases}$$

Find (i) the value of the constant c so that $f(x)$ is a PDF and (ii) the mean of X .

Problem 4 (Continuous random variables, expected value) A trader receives a bonus if the total losses X (in \$100,000) of his investment is less than 1.5. The bonus equals $0.05 - \frac{X}{30}$ if $X < 1.5$, and equals 0 otherwise. Suppose X has PDF $f(x) = \frac{3}{x^4}$ for $x > 1$ and $f(x) = 0$ for $x \leq 1$. Find the expected value of the bonus.

Hint: $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$, where $f(x)$ is the PDF of X .

What is exponential dist. ?



T is the time taken to the **next event**.

Recap: Exponential Dist.

We often let $\lambda = \frac{1}{\theta}$, and say that the random variable X has an **exponential distribution** with **scale parameter** θ , denoted by $X \sim \text{Exp}(\theta)$, if its PDF is defined by

$$f(x) = \frac{1}{\theta} e^{-x/\theta}, \quad 0 \leq x < \infty.$$

Theorem 10 (Exponential distribution)

If $W \sim \text{Exp}(\theta)$ with **scale parameter** θ , then

$$\mathbb{E}[W] = \theta, \quad \text{Var}[W] = \theta^2.$$

Equivalently, in terms of the **rate parameter** $\lambda = \frac{1}{\theta}$, we have

$$\mathbb{E}[W] = \frac{1}{\lambda}, \quad \text{Var}[W] = \frac{1}{\lambda^2}.$$

Recap: Uniform Dist.

Example 5 (Uniform distribution)

The random variable X has a **uniform distribution** if its PDF $f(x)$ is equal to a constant on its **support** (where $f(x)$ is not 0). In particular, if the support is the interval $[a, b]$, then

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

We denote it by $X \sim U(a, b)$.

Problem 5 (Exponential distribution)

- (a) A certain type of aluminium screen has, on the average, 3 flaws in a 100-foot roll.
 - (i) What is the probability that the first 40 feet in a roll contain no flaws?
 - (ii) What assumption did you make to solve part (i)?
- (b) Cars arrive at a tollbooth at a mean rate of 5 cars every 10 minutes according to a Poisson distribution. Let X be the time in minutes that the toll collector will have to wait before collecting the first toll. Find $\mathbb{E}[X]$ and the standard deviation of X .

Problem 6 (Uniform distribution)

- (a) Let X be uniformly distributed over the interval $[a, b]$, i.e. $X \sim U(a, b)$. Find $\mathbb{E}[X]$. & $\text{Var}(X)$.
- (b) Let X be uniformly distributed over the interval $[0, 1]$. Let $Y = X^2$. Find the CDF and hence the PDF of Y .

Hint: Differentiating CDF will give the PDF.

Problem 7 (Poisson distribution) A store selling newspapers orders only $n = 4$ of a certain newspaper because the manager does not get many requests for that publication. If the number of requests per day follows a Poisson distribution with mean 3, **Hint:**

(i) What is the expected value of the number sold per day?

How many can I sell if I have 5 customers that day given that I only have 4 newspaper?

(ii) What is the minimum number that the manager should order so that the chance of having more requests than available newspapers is less than 0.05?

You may use the CDF table for Poisson distribution attached at the end of this worksheet for calculations.

Hint: Let X be the number of requests per day, and Y be the number of newspapers sold per day. For part (i), we are interested in finding $\mathbb{E}[Y]$. How can we get the PMF for Y based on what we know about X ? Remember that the store only has 4 newspapers.

	$\lambda = E(X)$									
x	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0	0.905	0.819	0.741	0.670	0.607	0.549	0.497	0.449	0.407	0.368
1	0.995	0.982	0.963	0.938	0.910	0.878	0.844	0.809	0.772	0.736
2	1.000	0.999	0.996	0.992	0.986	0.977	0.966	0.953	0.937	0.920
3	1.000	1.000	1.000	0.999	0.998	0.997	0.994	0.991	0.987	0.981
4	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999	0.998	0.996
5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999
6	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
x	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
0	0.333	0.301	0.273	0.247	0.223	0.202	0.183	0.165	0.150	0.135
1	0.699	0.663	0.627	0.592	0.558	0.525	0.493	0.463	0.434	0.406
2	0.900	0.879	0.857	0.833	0.809	0.783	0.757	0.731	0.704	0.677
3	0.974	0.966	0.957	0.946	0.934	0.921	0.907	0.891	0.875	0.857
4	0.995	0.992	0.989	0.986	0.981	0.976	0.970	0.964	0.956	0.947
5	0.999	0.998	0.998	0.997	0.996	0.994	0.992	0.990	0.987	0.983
6	1.000	1.000	1.000	0.999	0.999	0.999	0.998	0.997	0.997	0.995
7	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999	0.999
8	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
x	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8	4.0
0	0.111	0.091	0.074	0.061	0.050	0.041	0.033	0.027	0.022	0.018
1	0.355	0.308	0.267	0.231	0.199	0.171	0.147	0.126	0.107	0.092
2	0.623	0.570	0.518	0.469	0.423	0.380	0.340	0.303	0.269	0.238
3	0.819	0.779	0.736	0.692	0.647	0.603	0.558	0.515	0.473	0.433
4	0.928	0.904	0.877	0.848	0.815	0.781	0.744	0.706	0.668	0.629
5	0.975	0.964	0.951	0.935	0.916	0.895	0.871	0.844	0.816	0.785
6	0.993	0.988	0.983	0.976	0.966	0.955	0.942	0.927	0.909	0.889
7	0.998	0.997	0.995	0.992	0.988	0.983	0.977	0.969	0.960	0.949
8	1.000	0.999	0.999	0.998	0.996	0.994	0.992	0.988	0.984	0.979
9	1.000	1.000	1.000	0.999	0.999	0.998	0.997	0.996	0.994	0.992
10	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999	0.998	0.997
11	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999
12	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

x	3.0
0	0.050
1	0.199
2	0.423
3	0.647
4	0.815
5	0.916
6	0.966
7	0.988
8	0.996
9	0.999
10	1.000
11	1.000
12	1.000

Recap: Gamma

Let W denote the **waiting time until the α th occurrence** in a Poisson process with $\lambda = \frac{1}{\theta}$. Then W has a **Gamma distribution** with **shape** parameter α and **scale** parameter θ , denoted by $W \sim \text{Gamma}(\alpha, \theta)$, with PDF given by

$$f(w) = \frac{1}{\Gamma(\alpha)\theta^\alpha} w^{\alpha-1} e^{-w/\theta}, \quad 0 \leq w < \infty.$$

Here:

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy, \quad \alpha > 0.$$

For our purpose, α is usually a positive integer, and so

$$\Gamma(\alpha) = (\alpha - 1)!.$$

Theorem 12 (Gamma distribution)

If $W \sim \text{Gamma}(\alpha, \theta)$, then

$$\mathbb{E}[W] = \alpha\theta, \quad \text{Var}[W] = \alpha\theta^2.$$

Deriving Gamma dist.

We have learnt that the waiting time until the **first phone call** follows **exponential** distribution.

Generalisation: How about the waiting time T_n until the **n -th** phone call?

Let $F(t)$ be the **CDF** of T_n , then

$$F(t) = \mathbb{P}(T_n < t) = \mathbb{P}(\text{at least } n \text{ calls in } [0, t])$$

Poisson

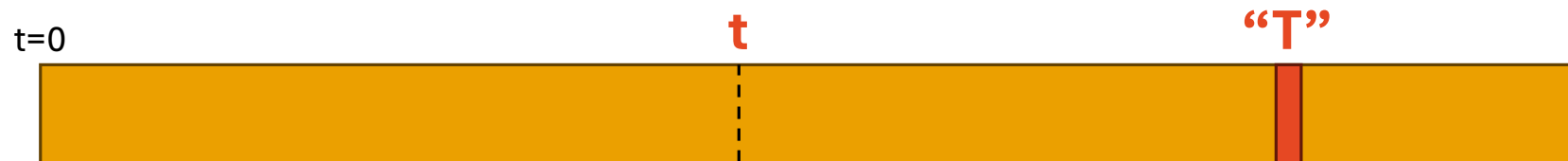
$$= \sum_{k=n}^{\infty} \frac{(\lambda t)^k e^{-\lambda t}}{k!} = 1 - \sum_{k=0}^{n-1} \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

hence the probability density function is

$$p(t) = F'(t) = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}.$$

Where $\theta = 1/\lambda$.

Exponential, Poisson and Gamma – Are they related?



Consider a degenerate case where we are waiting for **1 event** to occur.

$W \sim \text{Gamma}(\alpha = 1, \theta)$

$$f(w) = \frac{1}{\Gamma(\alpha)\theta^\alpha} w^{\alpha-1} e^{-w/\theta}$$

$$= 1/\theta \exp(-w/\theta)$$

Hence, $W \sim \exp(\theta)$

Consider a Poisson Process or Poisson Distribution with rate $= \lambda t$.

$X \sim \text{Poisson}(\lambda t)$

$$p(x) = e^{-\lambda} \frac{\lambda^x}{x!},$$

Suppose **T** is the waiting time until the first event occurs. If the first event hasn't happened by time t , then clearly zero events occurred in the interval $[0, t]$.

$$\Pr(T > t) = \Pr(X = 0) = \exp(-\lambda t)$$

$$\Pr(T \leq t) = 1 - \Pr(T > t) = 1 - \exp(-\lambda t)$$

Which is the CDF of **$\exp(\theta = 1/\lambda)$** .

Hence, **$T \sim \exp(\theta = 1/\lambda)$** .

2. Relation between the Poisson and exponential distributions

An interesting feature of these two distributions is that, if the Poisson provides an appropriate description of the number of occurrences per interval of time, then the exponential will provide a description of the length of time between occurrences. To understand this, consider that, in a Poisson process, if events occur on average at the rate of λ per unit of time, then there will be on average λt occurrences per t units of time. The Poisson distribution describing this process is therefore $P(x) = e^{-\lambda t} (\lambda t)^x / x!$, from which $P(x = 0) = e^{-\lambda t}$ is the probability of no occurrences in t units of time.

Another interpretation of $P(x = 0) = e^{-\lambda t}$ is that this is the probability that the time, T , to the first occurrence is greater than t , i.e.

$$P(T > t) = P(x = 0 \mid \mu = \lambda t) = e^{-\lambda t}.$$

Conversely, the probability that an event does occur during t units of time is given by

$$P(T \leq t) = 1 - P(x = 0 \mid \mu = \lambda t) = 1 - e^{-\lambda t}.$$

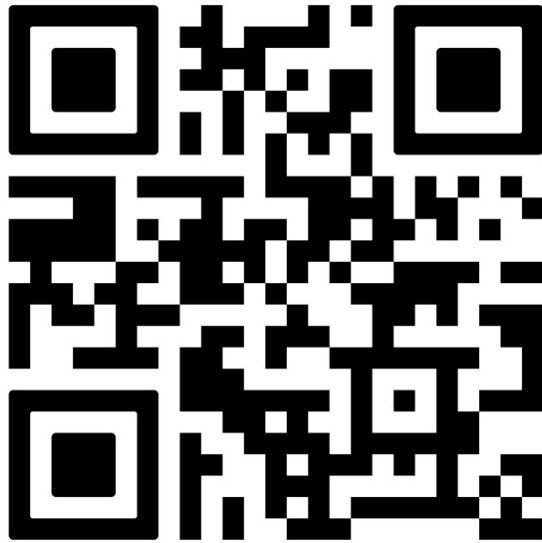
Note that this is the cumulative exponential distribution which, when differentiated with respect to t , produces the probability density function of the exponential distribution $f(t) = \lambda e^{-\lambda t}$.

References:

https://neuropsychics.ucsd.edu/courses/physics_171/exponential.pdf

Note: If you would like to learn more about this, you can take a BDE called **MH3512 Stochastic Process** (Only offered in Sem 1).

CNY: Tutorial 5



https://youtu.be/Fpz_t-1S0Qo



Zoom call for QnA

19 Feb (Thur); 8-9pm

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